Practical No.5

Aim: To understand AWS Lambda, its workflow, various functions and create your first Lambda functions using Python / Java / Nodejs.

Theory:

AWS Lambda is a serverless compute service from Amazon Web Services (AWS) that runs your code without the need to manage servers. You upload code, and Lambda takes care of running it, scaling automatically based on demand.

Key Concepts:

1. **Code Unit**: The script or program you upload to Lambda that gets executed.
2. **Event Source**: An external trigger, such as S3, DynamoDB, API Gateway, etc., that starts the execution of your code.
3. **Execution Role**: The permissions granted to Lambda to access other AWS resources, configured via an Identity and Access Management (IAM) role.
4. **Runtime**: The environment where your code executes. Various programming languages are supported, such as Python, Node.js, Java, and others.
5. **Concurrency**: The number of executions happening at the same time. Lambda scales automatically to handle multiple events.

Uses of Lambda function:

- **Data Processing**: Handle real-time data from AWS services like S3 or Kinesis.
- **Web Apps**: Act as the backend for serverless web applications.
- **Scheduled Tasks**: Automate tasks triggered by time-based events (e.g., cron jobs).
- **File Transformations**: Process files like images uploaded to an S3 bucket.
- **Security**: Enforce policies or automate remediation of issues in your AWS environment.

Implementation:

Step 1: Sign in to the AWS Management Console

1. Go to the AWS Management Console.

2. Sign in with your AWS account credentials.

Step 2: Navigate to AWS Lambda

1. In the AWS Management Console, find and select Lambda from the Services menu.

Step 3: Create a New Lambda Function

1. Click on Create function.

2. You will be presented with four options:

o Author from scratch: Create a function from

scratch.

o Use a blueprint: Start with a sample Lambda

function.

o Container image: Deploy a function defined as a

container image.

o Browse serverless app repository: Use a pre-built

serverless application.

For simplicity, select Author from scratch.

Step 4: Configure the Function

1. Function name: Enter a unique name for your Lambda

function (e.g., MyFirstLambda).

2. Runtime: Choose the runtime environment for your code

(e.g., Python 3.x, Node.js 14.x, etc.).

3. Role:

o If you dont have an existing IAM role, select Create a new role with basic Lambda permissions.

o If you have an existing IAM role, you can use that

by selecting Use an existing role.

Step 5: Write the Lambda Function Code

1. Once the function is created, you will be redirected to the

functions configuration page.

2. Scroll down to the Code section.

3. In the code editor, you can write your Lambda function

code. For example, if you are using Python:

```
def lambda_handler(event, context):
return {
‘statusCode’;: 200,
‘body’;: ‘Hello, AWS Lambda!’
}
```

4. The lambda_handler function is the entry point for your Lambda function. The event parameter contains the input data, and the context parameter provides runtime information.

Step 6: Test the Function

1. At the top of the function configuration page, click on

Test.

2. Create a new test event by choosing Create new test

event.

3. Provide a name for the test event (e.g., TestEvent), and

you can use the default event template.

4. Click Create and then click Test again to invoke the

function.

Step 7: Review the Execution Results

1. After testing, you can review the execution results in the

Execution result section.

2. If everything is set up correctly, you should see the

output {“statusCode&quot”: 200, “body”: “Hello,

AWS Lambda!”}.

Step 8: Deploy the Function (Optional)

1. If you want to deploy the Lambda function as part of a more complex architecture, consider using AWS services like API Gateway, S3, or Step Functions to trigger the

Lambda function.

Step 9: Monitor and Manage

1. AWS Lambda provides built-in monitoring through Amazon CloudWatch. You can view logs and performance metrics to manage and troubleshoot your Lambda function.